

Problem Set 2

Tate Mason

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Problem 1: Data Overview and Analysis

Part A: Data Summary

In Colombia, we see fewer overall observations in every sector:

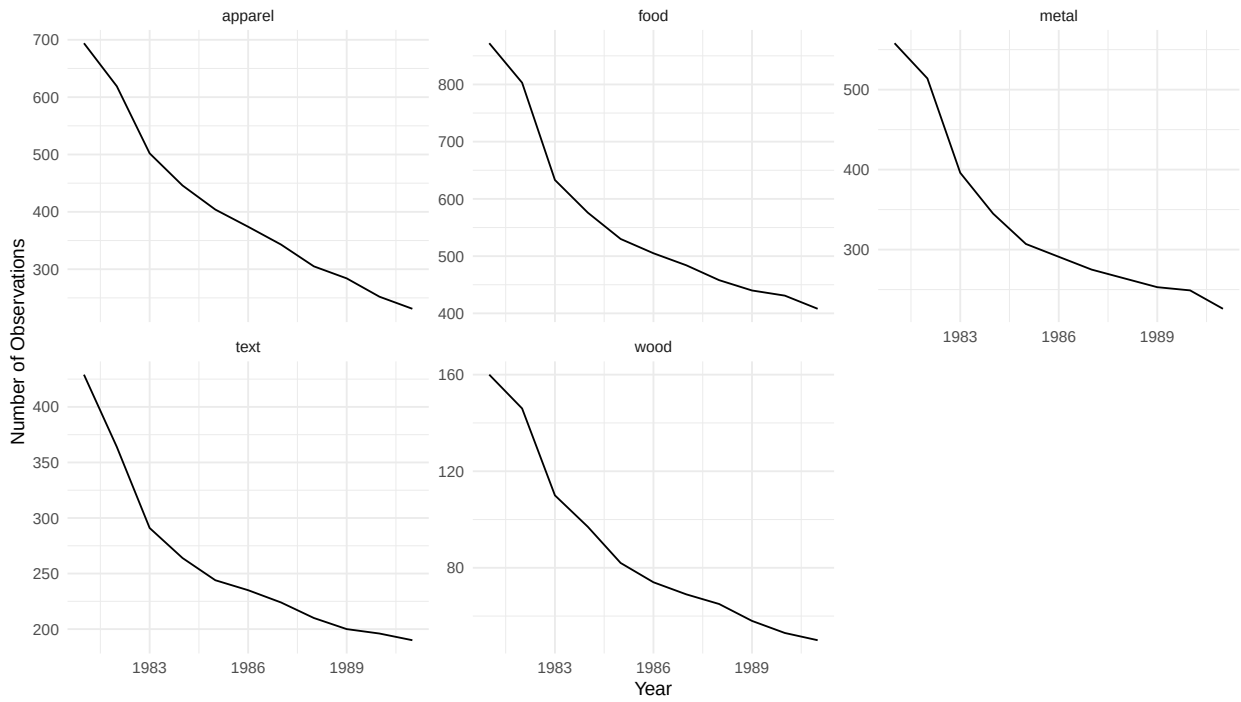
- Food Industry: 6140
- Apparel Industry: 4454
- Metal Industry: 3678
- Textile Industry: 2840
- Wood Industry: 964

In Chile, we see an interesting distribution of observations:

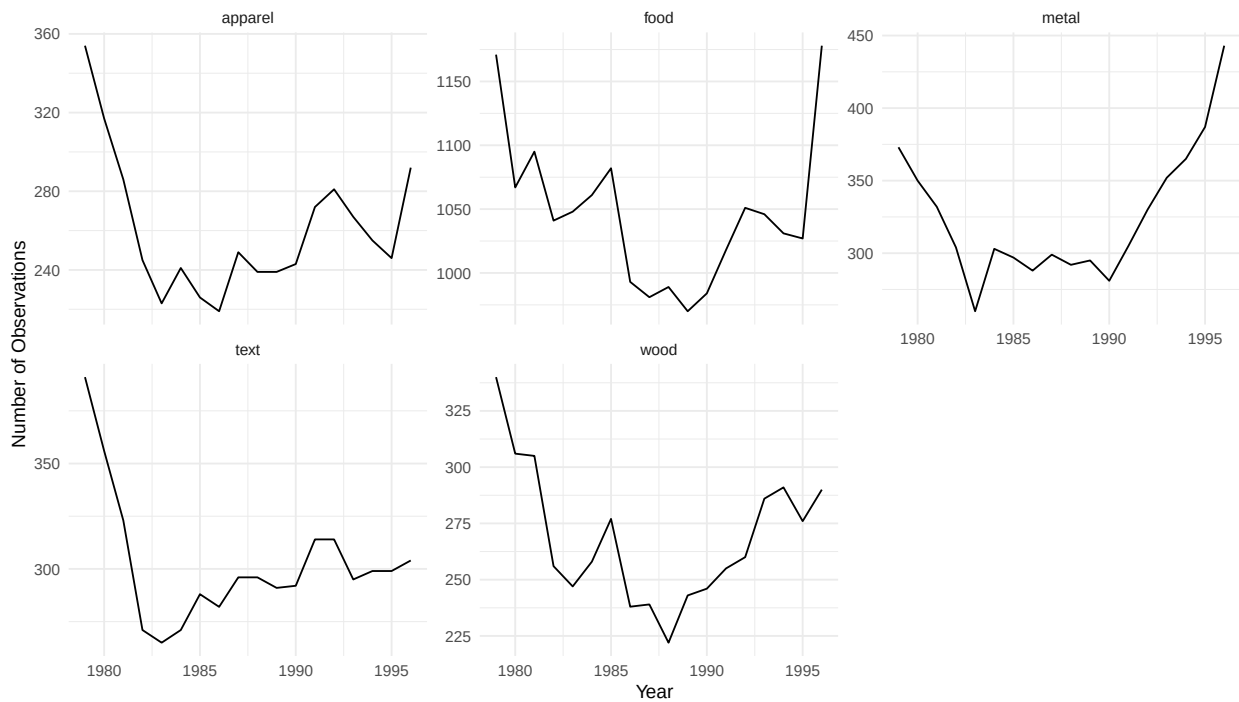
- Food Industry: 18833
- Metal Industry: 5856
- Textile Industry: 5447
- Wood Industry: 4835
- Apparel Industry: 4694

Both Chile and Colombia have massive food industries, in terms of observations. Chile, however, has more than three times the observations in Colombia. In Colombia, the apparel industry is the second largest, while in Chile, the metal industry is the second largest. The wood industry is the smallest in Colombia, while apparel is the smallest in Chile.

Colombia: Observations by Year and Industry



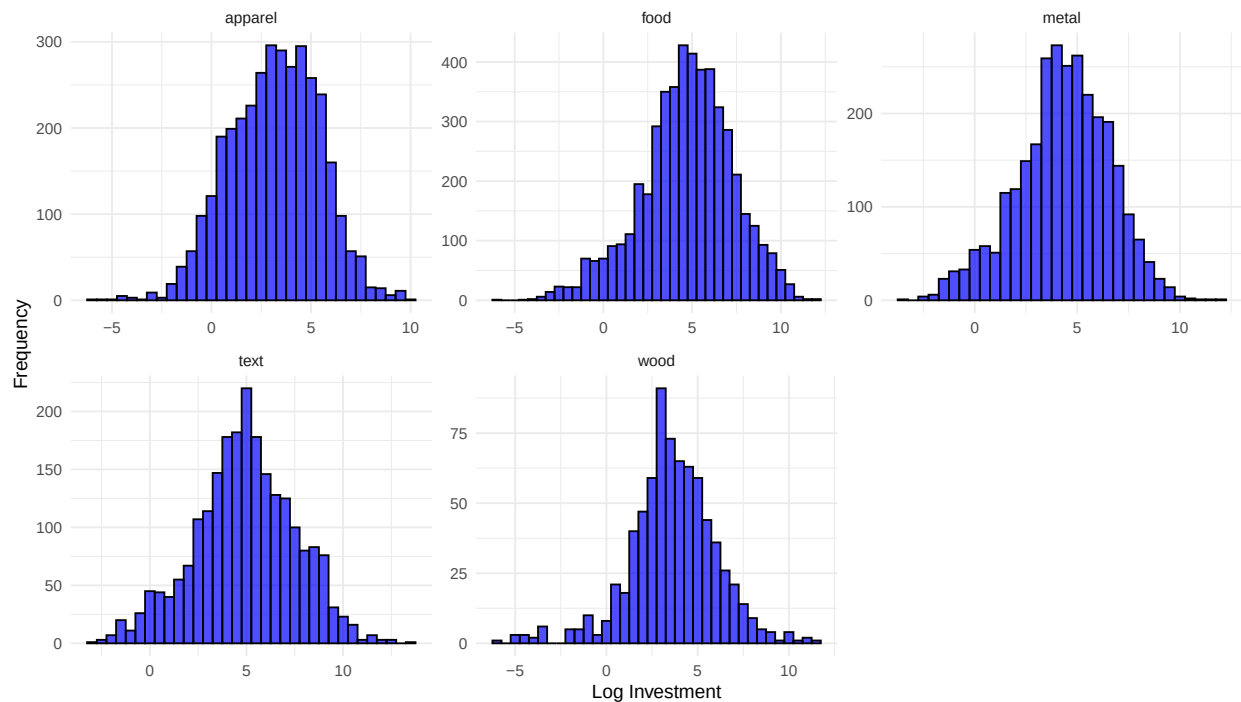
Chile: Observations by Year and Industry



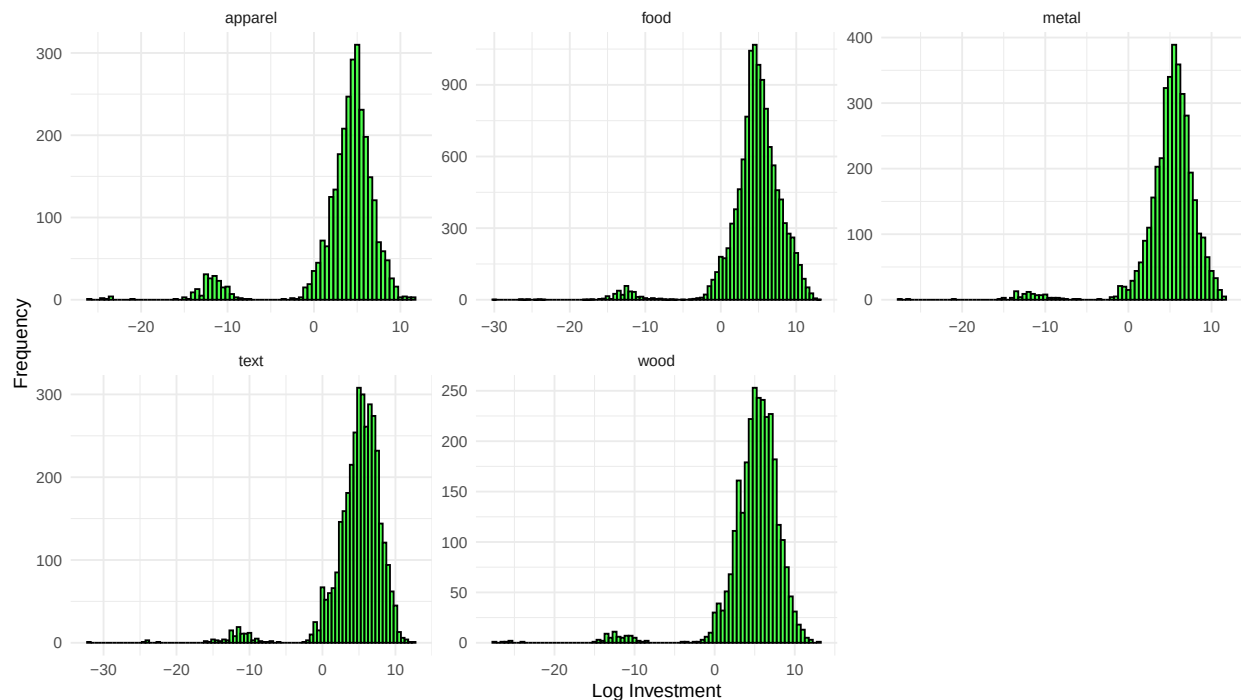
In Colombia, there appears to be a lack of data across the entire time span, with most data concentrated to the early 1980s. Thus, we see a massive decline in observations after 1981. In Chile, we see more diverse patterns in the data, with food and metal industries seeing a drop and then rebound, while apparel, textile, and wood industries see a more modest rebound from a valley in the late 1980s.

Part B: Analysis of Investment by Industry

Histogram of Log Investment by Industry, Colombia



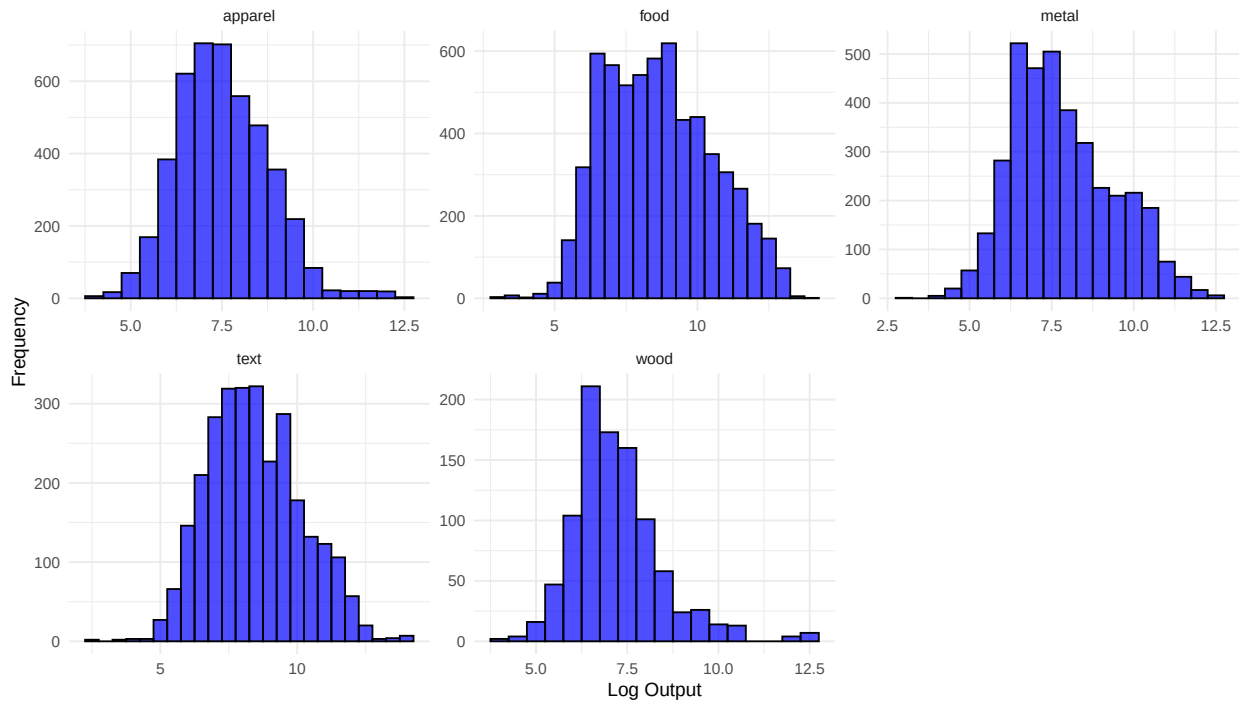
Histogram of Log Investment by Industry, Chile



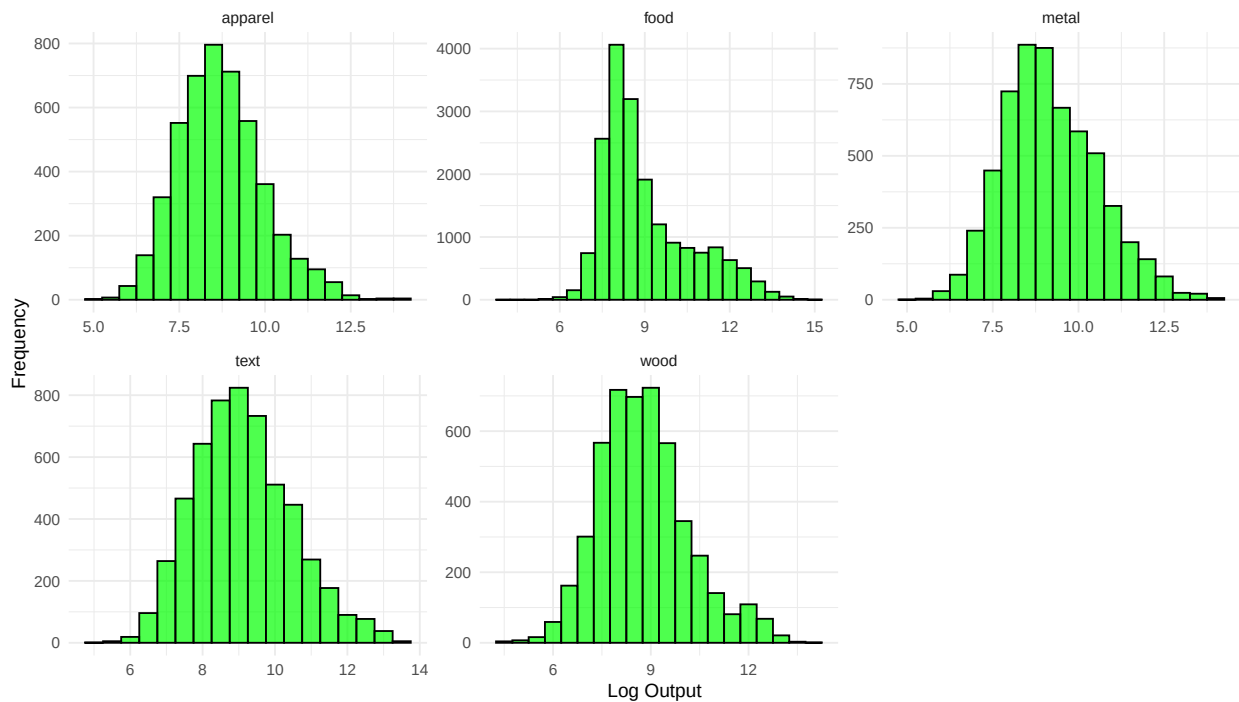
In Colombia, there is much more dispersion in investment levels across industries, with some industries even experiencing negative investment. Food, metal, and textiles have very similar investment patterns, with means around 5. Apparel and wood are similar to the other three, though a bit lower. In Chile, investment levels are super consistent across industries, with all industries having means around 5-7. Distributions are also much tighter, though most do have some negative investment observations.

Part C: Analysis of Ouput by Industry

Histogram of Log Output by Industry, Colombia



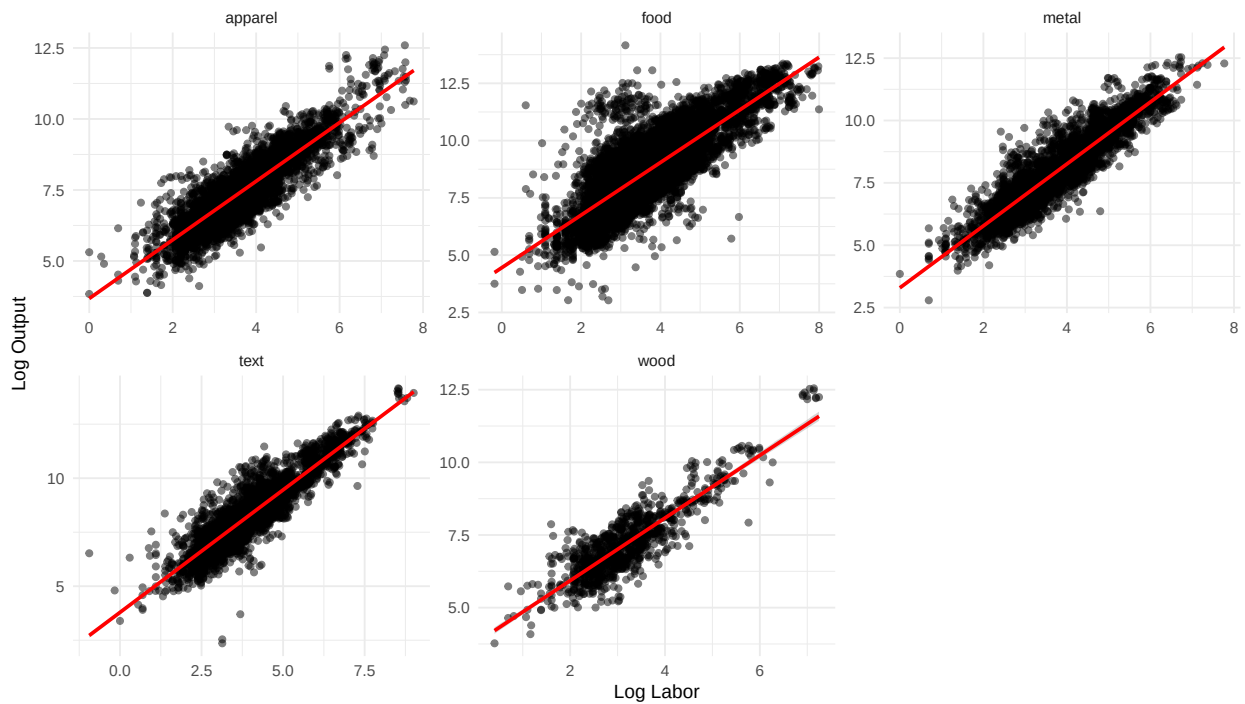
Histogram of Log Output by Industry, Chile



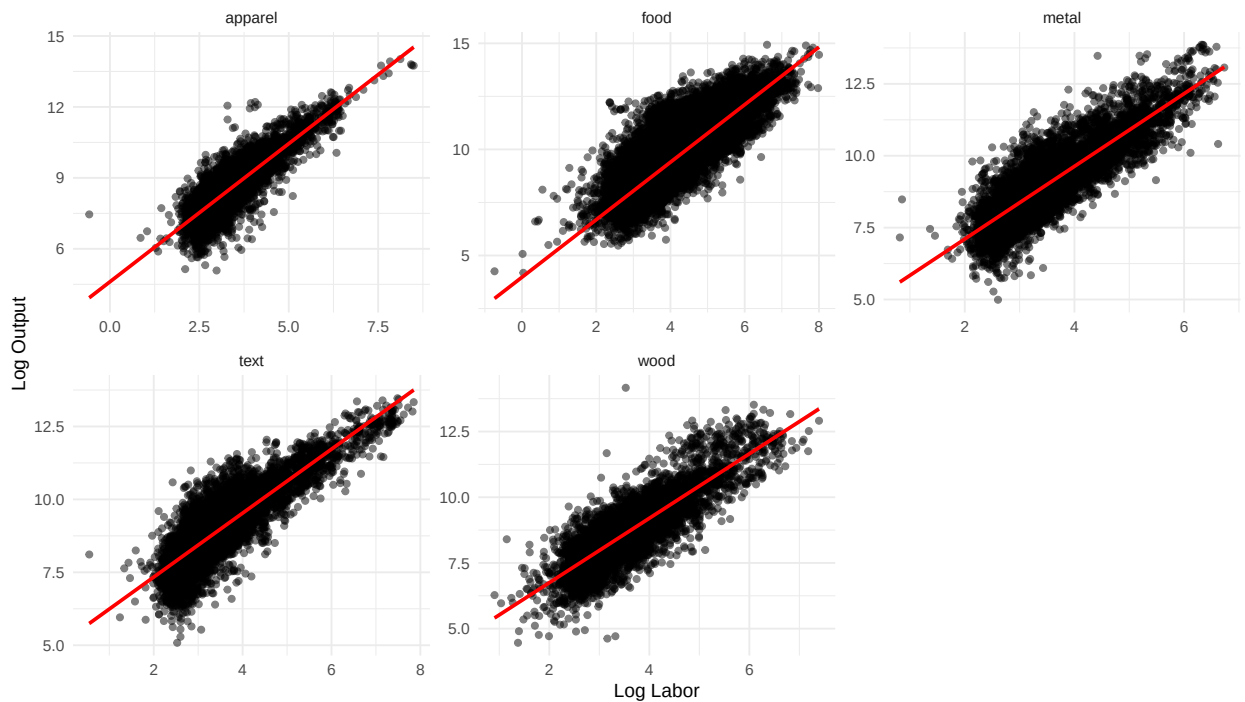
In Colombia, output levels do not vary too much across industries, with means around 7-10 and similar distributions. In Chile, distributions are also similar, though there is a bit more variation. Textiles and food have a wider distribution of output levels, while apparel and metal are a bit tighter.

Part D: Analysis of Labor Output Correlation by Industry

Scatter Plot of Log Output vs Log Labor (Colombia)



Scatter Plot of Log Output vs Log Labor (Chile)



In both countries, there is a positive relationship between labor input and output across all industries. The strength of this relationship varies by industry, with some industries showing a stronger correlation than others. For example, in Colombia, the food industry shows a strong positive correlation, while the wood industry has a weaker correlation. In Chile, all industries show a strong positive correlation.

Problem 2: Estimation Techniques

Comparison of Estimation Techniques

In OLS, we perform a simple linear regression of value-added output on inputs. This method yields very high estimates for labor elasticity in Colombia and high capital elasticity in Chile, suggesting that these inputs play a significant role in production. However, OLS does not account for potential endogeneity of inputs, which can bias the estimates. In the heterogeneous case, we see that while labor elasticity remains high in Colombia, Chile flips, and capital elasticity becomes smaller than labor elasticity across all industries.

When we add investment to our OLS regression, we see a slight decrease in capital and labor elasticities in both countries, with labor elasticity still being dominant in both countries. This suggests that investment may be capturing some of the effects previously attributed to capital and labor. In the heterogeneous case, we see similar patterns, with investment elasticity being small but positive across most industries, though not all.

The Olley-Pakes (OP) method addresses the endogeneity of inputs by using investment as a proxy for unobserved productivity shocks. This method yields lower estimates for both capital and labor elasticities, suggesting that previous OLS specifications may have been biased upwards. In the heterogeneous case, we see that capital elasticity is even lower across almost all industries, as is the coefficient on the productivity shock function, α_ϕ .

Levinsohn-Petrin (LP) further refines the estimation by using intermediate inputs as a proxy for productivity shocks. This method yields even lower estimates for capital and labor elasticities, with intermediate input elasticity being relatively large in both countries. This suggests that intermediate inputs play a large role in production, potentially capturing effects previously attributed to the capital and labor inputs. In the heterogeneous case, we see that intermediate input elasticity is mostly larger than in the homogenous case in Colombia, while in Chile it is mostly smaller. Comparing to OP, LP yields lower capital elasticities in all industries in Colombia as well as in Chile. This points to intermediate inputs being a more relevant proxy than investment for productivity shocks.

Finally, ACF builds on OP and LP, addressing simultaneity bias in input choices. This method yields higher estimates for both capital and labor elasticities compared to OP and LP, but still lower than OLS. This suggests that while ACF corrects for some bias, it still does not fully capture the whole effect of endogeneity. In the heterogeneous case, we see mostly higher labor elasticities in Colombia, with only food and textile industries having lower labor elasticities compared to the homogenous case. In Chile, all industries have higher labor elasticities compared to the homogenous case, except for textiles. For capital, the case is mixed, with apparel, metal, and wood having lower capital elasticities in Colombia, while this is the case for all but food in Chile. Comparing to OP and LP, ACF yields higher capital elasticities across industries, for the most part, in both countries. This suggests that ACF is able to better capture the true effect of capital on production.

Overall, it is obvious that the choice of estimation technique has a significant impact on the estimated production function parameters. Different methods yield varying estimates for capital, labor, and the other inputs considered. This underscores the importance of selecting an appropriate estimation method that adequately addresses endogeneity and simultaneity biases in input choices.

Part A: OLS

To estimate via OLS, we estimate the following equation:

$$Y_{it} = e^{\alpha_0 + \omega_{it} + \varepsilon_{it}} K_{it}^{\alpha_k} L_{it}^{\alpha_l} \quad (1)$$

$$y_{it} = \alpha_0 + \omega_{it} + \alpha_k k_{it} + \alpha_l l_{it} + \varepsilon_{it} \quad (2)$$

Where y is log of value-added output, k is log capital, l is log labor, and ω is the productivity shock. We estimate this equation for both Colombia and Chile, first homogenous across industries, then heterogeneous by industry.

OLS Homogenous Results:

Country	α_0	α_k	α_l	R^2
Colombia (Homogenous)	2.186	0.273	0.831	0.823
Chile (Homogenous)	2.351	0.887	0.303	0.660

OLS Heterogenous Results:

Country	Industry	α_0	α_k	α_l	R^2
Colombia					
	Food	2.238	0.321	0.754	0.831
	Apparel	2.510	0.133	0.921	0.831
	Metal	1.974	0.208	0.998	0.831
	Textile	2.316	0.257	0.858	0.831
	Wood	2.559	0.117	0.858	0.831
Chile					
	Food	1.950	0.334	0.896	0.689
	Apparel	2.865	0.150	1.055	0.689
	Metal	2.734	0.232	0.994	0.689
	Textile	2.709	0.243	0.874	0.689
	Wood	2.488	0.174	1.016	0.689

These results indicate that there are significant differences in the production function parameters across countries and industries. Colombia has a higher labor elasticity compared to Chile, suggesting that labor plays a more important role in production in Colombia. Conversely, Chile has a higher capital elasticity, indicating that capital is more important in their production process. When looking at heterogenous estimates, we see that within each country, different industries have varying elasticities, reflecting the unique production processes and input requirements of each industry.

Part B: OLS with Investment

To estimate via OLS with investment, we modify our previous equation to include investment as an additional regressor:

$$y_{it} = \alpha_0 + \omega_{it} + \alpha_k k_{it} + \alpha_l l_{it} + \alpha_i i_{it} + \varepsilon_{it} \quad (3)$$

Where i is log investment. We estimate this equation for both Colombia and Chile, first homogenous across industries, then heterogenous by industry.

OLS with Investment Homogenous Results:

Country	α_0	α_k	α_l	α_i	R^2
Colombia (Homogenous)	2.282	0.242	0.828	0.031	0.836
Chile (Homogenous)	2.521	0.299	0.844	0.017	0.689

OLS with Investment Heterogenous Results:

Country	Industry	α_0	α_k	α_l	α_i	R^2
Colombia						
	Food	2.384	0.267	0.767	0.042	0.844
	Apparel	2.545	0.119	0.925	0.015	0.844
	Metal	2.041	0.196	0.985	0.017	0.844
	Textile	2.278	0.230	0.834	0.045	0.844
	Wood	2.518	0.139	0.954	-0.001	0.844
Chile						
	Food	2.091	0.339	0.843	0.015	0.715
	Apparel	3.106	0.146	1.006	0.016	0.715
	Metal	3.207	0.218	0.943	0.022	0.715
	Textile	2.359	0.202	0.846	0.027	0.715
	Wood	2.624	0.176	0.977	0.020	0.715

Part C: OP Estimation

Now, when we estimate with the Olley-Pakes method, we follow the two-step procedure outlined in the literature. In the first step, we estimate the production function parameters using a non-parametric approach to control for the endogeneity of input choices. In the second step, we use the estimated parameters to recover the productivity shocks.

That is, we estimate the following equation:

$$y_{it} = \alpha_l l_{it} + \phi(i_{it}, k_{it}) + \varepsilon_{it} \quad (4)$$

Where $\phi = \alpha_0 + \alpha_k k_{it} + \alpha_i i_{it} + h(i_t, k_{it})$. We estimate this equation for both Colombia and Chile, first homogenous across industries, then heterogenous by industry. We estimate via a second stage regression to recover α_k and α_ϕ :

$$y_{t+1} + \alpha_l l_{t+1} = \alpha_k k_{t+1} + \alpha_i i_{t+1} g(\phi_t - \alpha_i i_t - \alpha_k k_t - \alpha_0) + \varepsilon_{t+1} \quad (5)$$

Country	α_0	α_k	α_l	α_ϕ
Colombia (Homogenous)	0.839	0.218	0.776	0.510
Chile (Homogenous)	0.755	0.440	0.786	0.522

Now, for heterogenous estimates:

Country	Industry	α_0	α_k	α_ϕ
Colombia				
	Food	1.944	0.279	0.092
	Apparel	1.947	0.093	0.214
	Metal	1.227	0.165	0.309
	Textile	1.807	0.262	0.122
	Wood	1.746	0.058	0.288
Chile				
	Food	1.356	0.357	0.154
	Apparel	2.053	0.064	0.337
	Metal	2.480	0.218	0.228
	Textile	2.936	0.235	0.080
	Wood	1.759	0.152	0.263

Part D: LP Estimation

To estimate via Levinsohn-Petrin, we modify our previous equation to include intermediate inputs as a proxy for unobserved productivity shocks:

$$y_{it} = \alpha_l l_{it} + \phi(m_{it}, k_{it}) + \varepsilon_{it} \quad (6)$$

Where m is log intermediate inputs. We estimate this equation for both Colombia and Chile, first homogenous across industries, then heterogenous by industry.

We then estimate via a second stage regression to recover α_k and α_ϕ :

$$y_{t+1} + \alpha_l l_{t+1} = \alpha_k k_{t+1} + \alpha_m m_{t+1} g(\phi_t - \alpha_m m_t - \alpha_k k_t - \alpha_0) + \varepsilon_{t+1} \quad (7)$$

Country	α_0	α_k	α_l	α_m	α_ϕ
Colombia (Homogenous)	1.131	0.083	0.616	0.350	0.112
Chile (Homogenous)	0.673	0.0186	0.481	0.396	0.203

Now, for heterogenous estimates:

Country	Industry	α_0	α_k	α_m	α_ϕ
Colombia					
	Food	0.976	0.074	0.412	0.010
	Apparel	1.403	0.029	0.267	0.216
	Metal	1.040	0.067	0.434	0.037
	Textile	1.086	0.118	0.451	-0.143
	Wood	1.309	0.047	0.237	0.275
Chile					
	Food	0.014	0.155	0.495	0.177
	Apparel	1.153	0.009	0.221	0.518
	Metal	1.456	0.113	0.372	0.308
	Textile	1.536	0.136	0.275	0.308
	Wood	0.648	0.075	0.441	0.244

Part E: ACF Estimation

To estimate via Akerberg-Caves-Frazer, we follow a similar two-step procedure as in the OP method, but we also account for the simultaneity bias in input choices. In the first step, we now include intermediate inputs in the non-parametric function, ϕ , for estimation:

$$y_{it} = \alpha_0 + \alpha_k k_{it} + \alpha_l l_{it} + \alpha_m m_{it} + f^{-1}(m_{it}, k_{it}, l_{it}) + \varepsilon_{it} \quad (8)$$

Which can be used to recover productivity shocks as:

$$y_{it} = \phi(m_{it}, k_{it}, l_{it}) + \varepsilon_{it} \quad (9)$$

The second stage is now different, using lagged inputs as instruments to control for simultaneity bias:

$$y_t = \alpha_k k_t + \alpha_l l_t + g(\phi_{t-1} - \alpha_k k_{t-1} - \alpha_m m_{t-1} - \alpha_l l_{t-1} - \alpha_0) + \varepsilon_t \quad (10)$$

Country	α_0	α_k	α_l	α_ϕ
Colombia (Homogenous)	1.473	0.317	0.814	0.244
Chile (Homogenous)	1.525	0.344	0.885	0.249

Now, for heterogenous estimates:

Country	Industry	α_0	α_k	α_l	α_ϕ
Colombia					
	Food	1.200	0.402	0.677	0.352
	Apparel	2.224	0.140	0.921	0.107
	Metal	1.436	0.263	0.970	0.178
	Textile	1.702	0.321	0.812	0.131
	Wood	2.046	0.109	1.057	0.098
Chile					
	Food	1.068	0.383	0.893	0.289
	Apparel	2.268	0.149	1.086	0.149
	Metal	2.395	0.237	1.013	0.210
	Textile	2.232	0.278	0.858	0.219
	Wood	1.948	0.181	1.042	0.140

Problem 3: Productivity Dynamics

Part A: Estimation of ω Using OP Estimates - Homogenous

Country	Mean of ω	Std. Dev. of ω
Colombia	294.294	702.900
Chile	74.876	82.639

Colombia possesses much higher mean productivity than Chile, more than quadruple the mean. However, Colombia also has a much higher standard deviation, indicating significant disparities in productivity among firms. In fact, this standard deviation is so large that it suggests the presence of extreme outliers in the Colombian data. This warrants further looking into how Colombian productivity is distributed to better understand the underlying dynamics.

Part B: Estimation of ω Using OP Estimates - Heterogenous

Country	Industry	Mean of ω	Std. Dev. of ω
Colombia			
	Food	61.7	115
	Apparel	163	376
	Metal	238	400
	Textile	124	279
	Wood	294	1310
Chile			
	Food	90.9	124
	Apparel	460	1849
	Metal	88.7	119
	Textile	62.7	134
	Wood	357	619

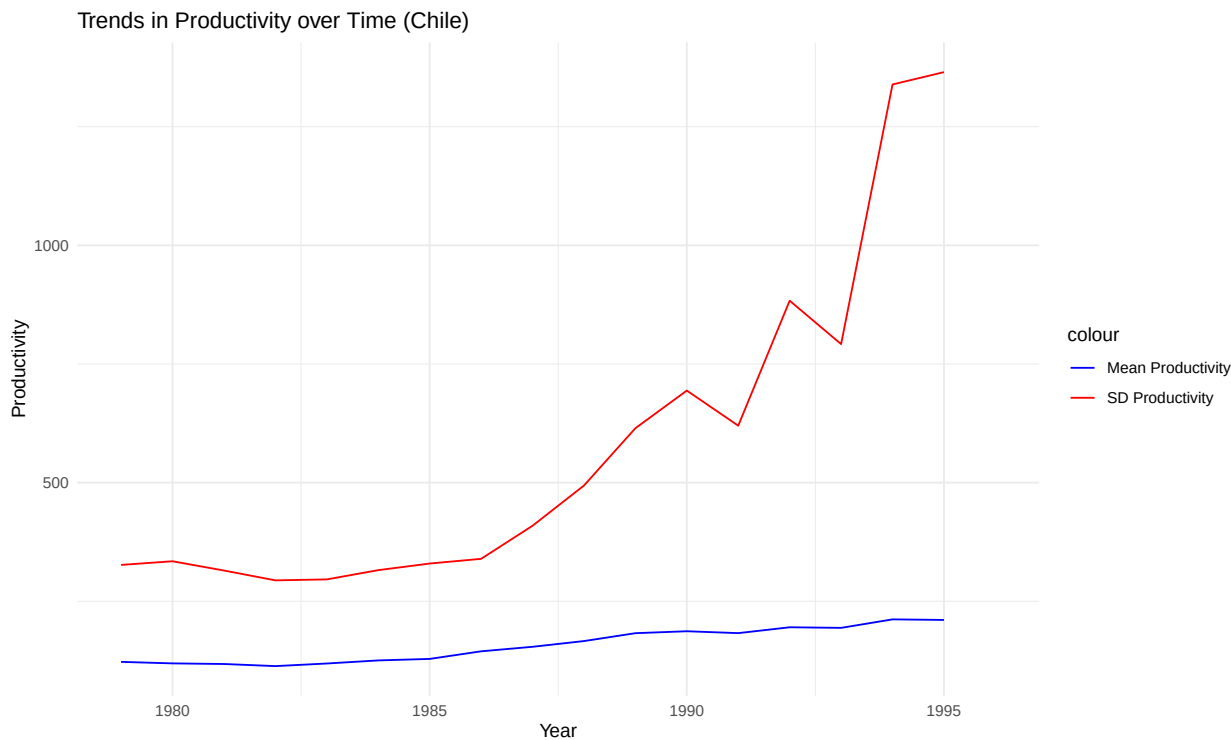
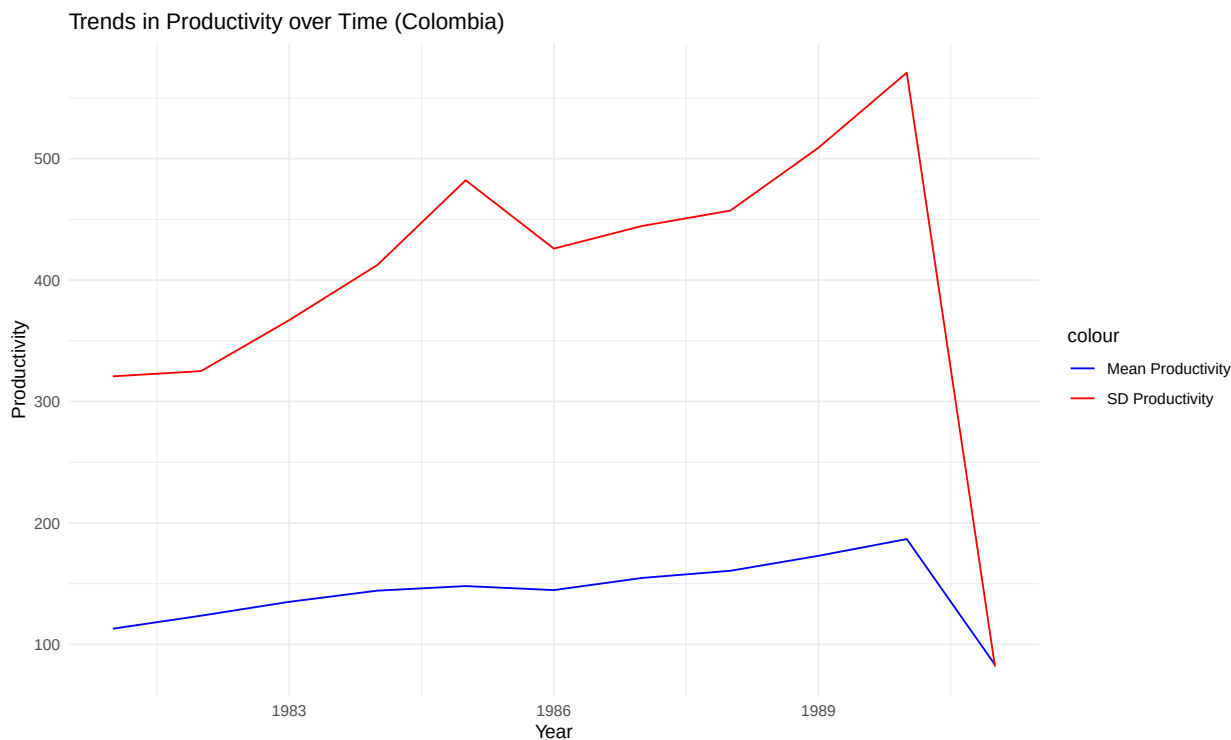
These are super weird results due to the massive standard deviations, likely caused by outliers in the data. What can be seen is that Colombia has a generally higher mean productivity than Chile across all industries except wood and apparel. Within Colombia, wood and metal industries have the highest mean productivity, while food has the lowest. In Chile, apparel and wood have the highest mean productivity, while textiles and metal have the lowest.

However, when looking at the output graphs for Colombia in 1.C, we actually see Wood and Metal firms having lower mean log output than other industries, despite having the highest mean productivity here. This suggests that these industries may have higher productivity but lower output levels, possibly due to smaller scale of operations or other factors limiting output. When looking at Chile's output graphs, we see that apparel and food do not have the highest mean log outputs, despite having the highest mean productivity

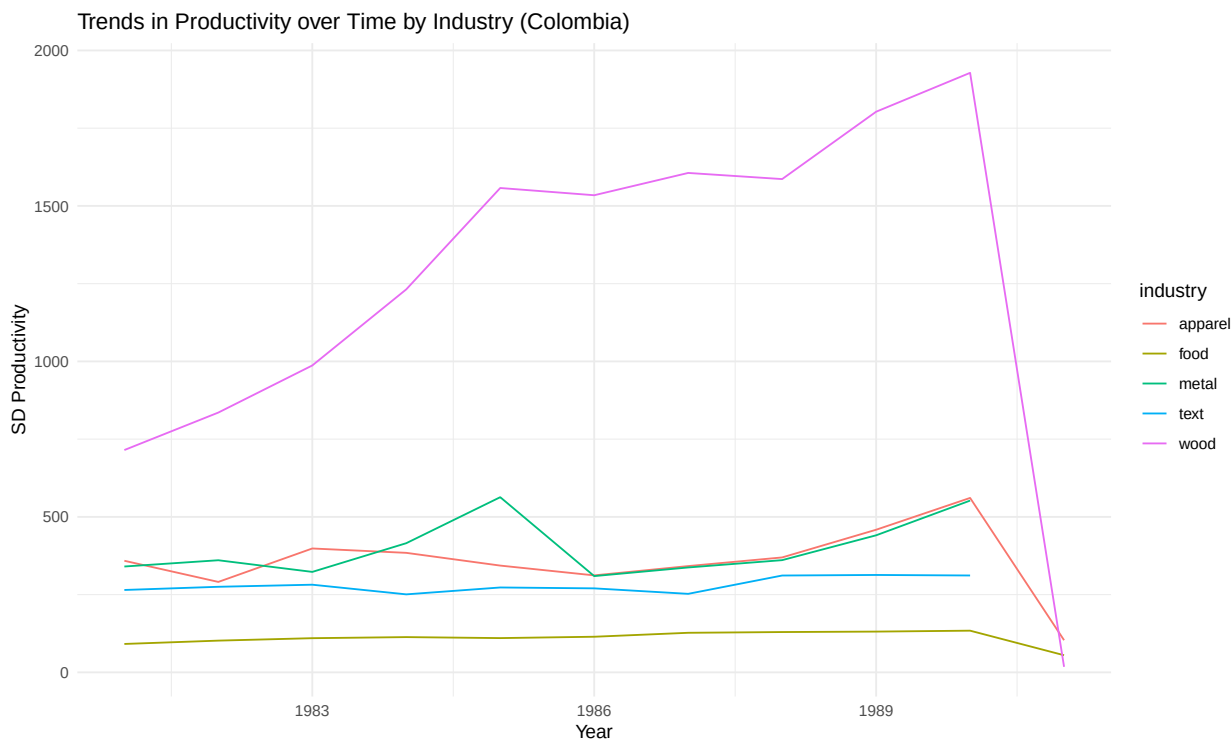
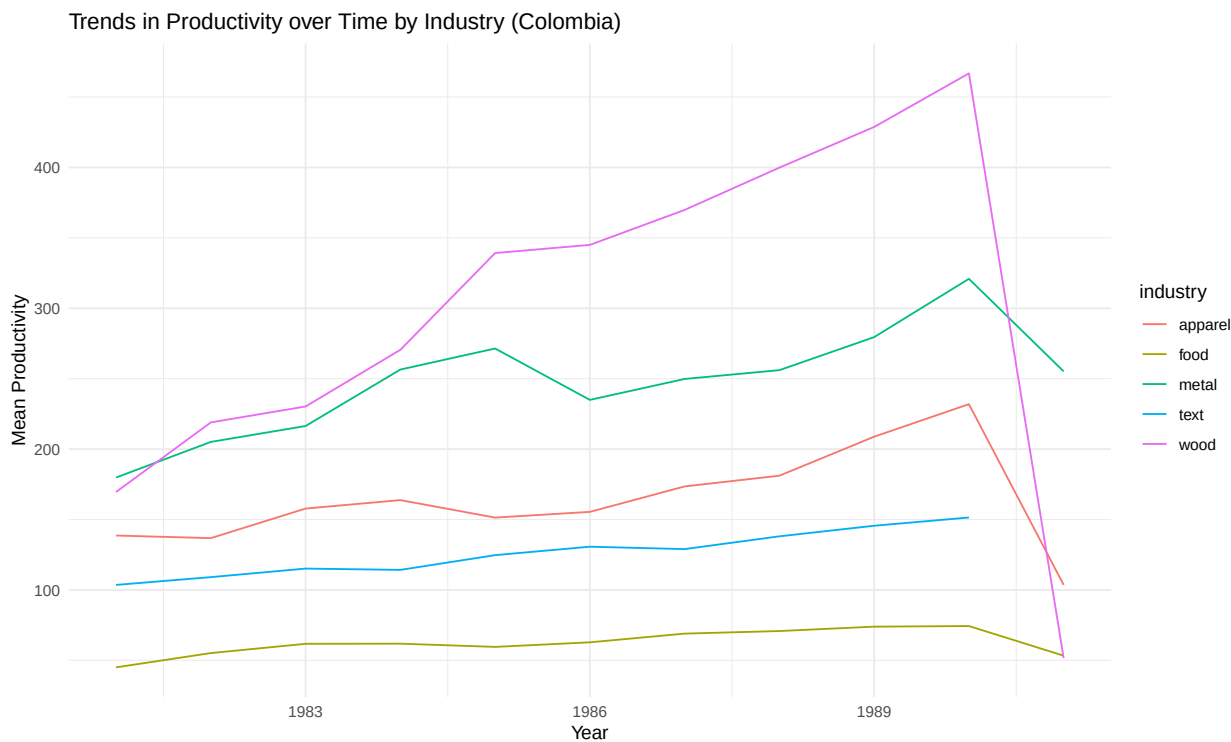
here. This again suggests that higher productivity does not necessarily translate to higher output levels, likely due to other limiting factors.

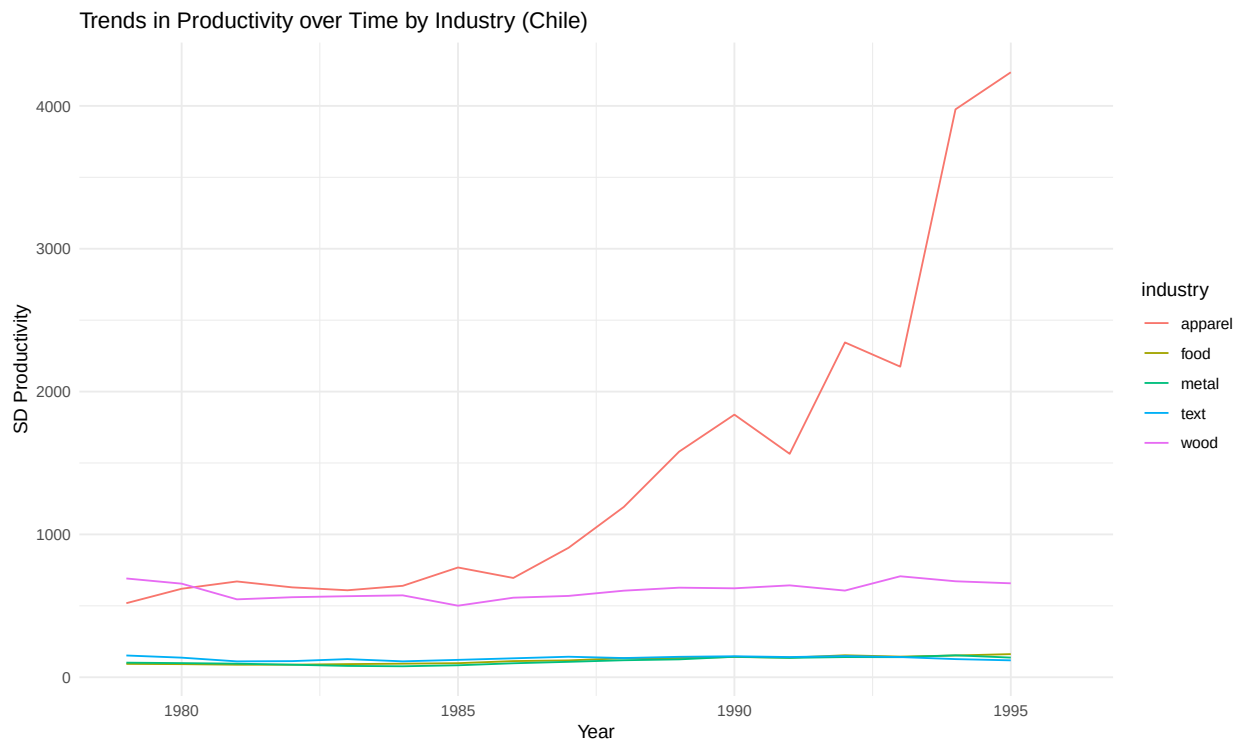
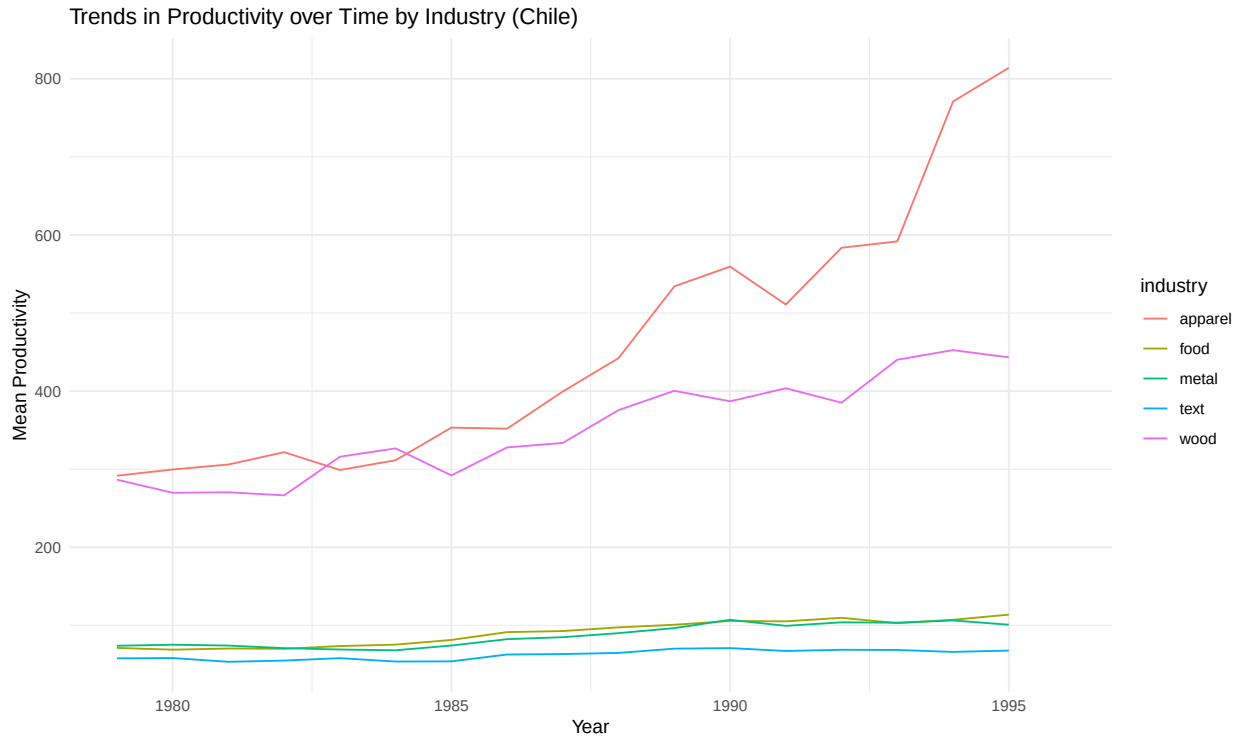
Part C: Time Trends in Productivity

Homogenous Productivity Trends



Heterogenous Productivity Trends by Industry





Discussion

In both countries, we observe growth in productivity over time, with increasing standard deviations as well, indicating growing disparities among firms. Homogenous productivity trends show a steady, moderate increase in productivity for both countries, with standard deviations growing during the same period.

Heterogeneous productivity trends show interesting patterns for different industries. In Colombia, The

food industry is the most productive industry with the highest growth in productivity over time. However, it also possesses the highest standard deviation, indicating significant disparities among firms. The other industries also show growth in productivity, but at a slower rate. The other industries also have lower standard deviations, which mostly do not grow, indicating more uniform productivity levels among firms.

In Chile, the apparel and food industries are the most productive, with apparel having the highest growth rate. However, apparel also has the highest standard deviation, indicating significant disparities among firms. The other industries show almost no growth in productivity, with almost no changes in standard deviations as well, indicating uniform productivity levels among firms.